

Assessment Guidelines

- This assessment must be completed individually without assistance from others.
- The use of AI-powered tools or code generation platforms (such as ChatGPT, GitHub Copilot, Gemini, or similar services) is not permitted.
- Please ensure that all responses are based on your own understanding, practical knowledge, and testing approach.
- Submissions that appear to be copied, AI-generated, or rely on generic templates may not be considered for further evaluation.
- Provide clear, well-structured, and original answers wherever applicable.

1. End-to-End Validation of Asynchronous File Processing

A financial platform enables users to upload transaction files in CSV format. Once uploaded, the files are processed asynchronously, and the data is synchronized with the application's internal records.

Describe how you would automate the complete workflow. Your answer should cover:

- Validation of the file upload through the UI
 - Verification of processing status using APIs or backend services
 - Handling and validating scenarios where only some records fail while others are processed successfully
 - Techniques for reliably validating asynchronous workflows without relying on fixed wait times
 - How you would ensure the overall data integrity from upload to completion
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2. Date Conversion and Validation

Create a solution that accepts a date in **MM/DD/YYYY** format and converts it to **YYYY-MM-DD**.

Additionally, explain how your solution would:

- Detect and reject invalid dates (for example, invalid days, months, or leap-year cases)
 - Identify whether the given date falls on a weekday or a weekend
 - Handle unexpected or incorrectly formatted inputs
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3. API Performance Monitoring Strategy

In enterprise applications, response times may occasionally increase because of dependent services or network latency. Rather than failing a test due to a single slow response, design an approach that identifies recurring performance degradation.

Your explanation should include:

- How you would measure and track API response times
- Criteria for distinguishing temporary latency from a consistent performance issue
- Thresholds or statistical techniques you would use
- How this strategy can be incorporated into a Continuous Integration (CI) pipeline to minimize false positives while still detecting genuine performance regressions.